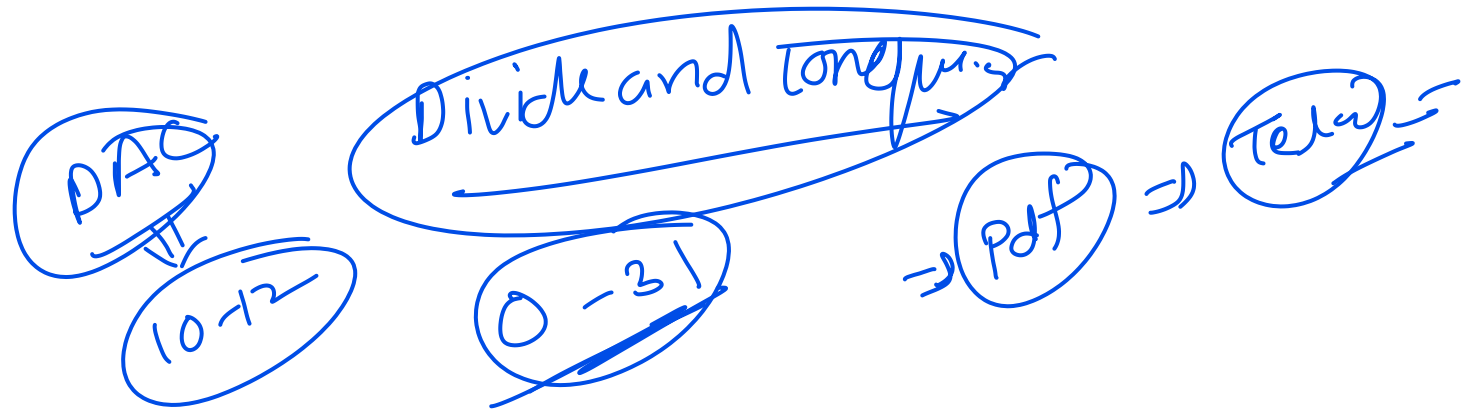




Lec_32

① ⇒ min-max

2. Power of an Element

$$3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \Rightarrow \text{power of } 3, \text{ thing}$$

$$4^3 = 4 \times 4 \times 4$$

Input: an element a , with exponent n
output: a^n .

- Without DAC:

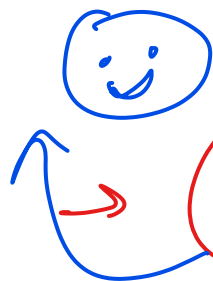
$$a^n = \underbrace{a \times a \times a \times a \dots}_{n \text{ times}}$$

```
Ans = 1
for (i = 1; i <= n; i++)
{
    Ans = Ans * a
}
```

$\Rightarrow O(n)$
execute n times

$\Rightarrow$ can be do in $\log n$ times

$$\log n \ll n$$



Time

$\Rightarrow$

$\gg \gg \gg$

space

actual exponent

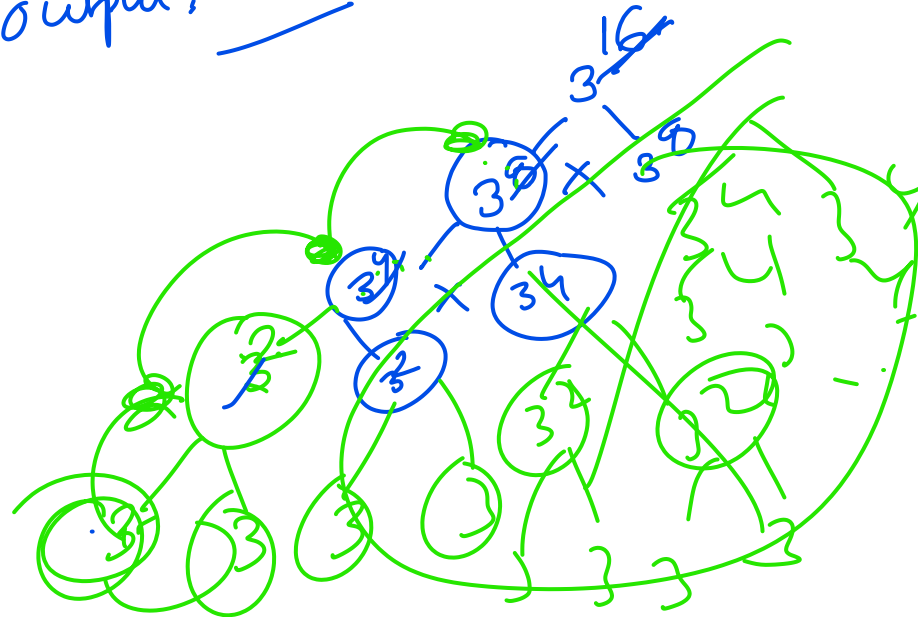
input: a, n
 output: a^n

$$3^4 \cdot 3^4 = 3^8$$

$$a^n$$

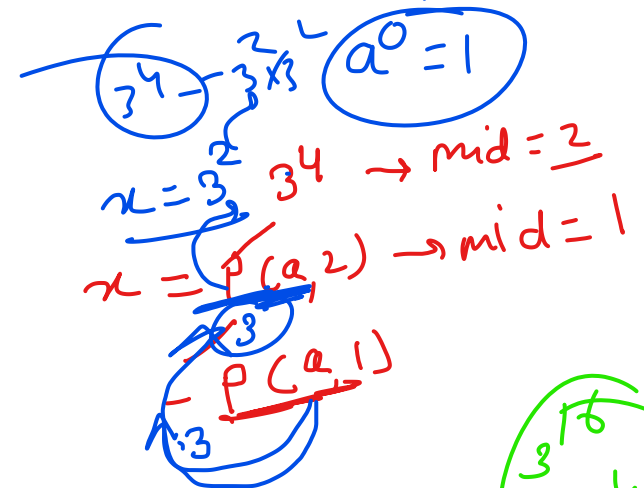
With DAC:

- Algo:



$$a^n = ?$$

if $n == 0, n == 1$

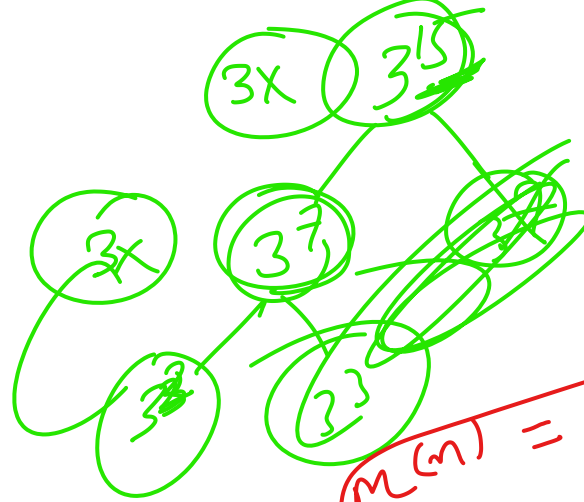


$$3^4$$

if n is
 index
 from of
 power of
 2

power(a, n)
 { if ($n == 0$) { return 1; }
 if ($n == 1$) { return a ; }
 else { mid = $n/2$
 $n =$ power(a, mid)
 return $n * n$; }

#



$$\frac{15}{2} = 7$$

$$T_2 = 3$$

$$T(n) = T(n/2) + 2$$

for all cases of n

power(a, n)

if (n == 0) { return 1; }

if (n == 1) { return a; }

else

mid = n/2

n = power(a, mid)

if (n * 2 == 0)

return n

else return (a * n * n)



$$T_c = ?$$

$$T(n) = T(n/2) + 1$$

$\bullet$ $TC = ?$

$na \Rightarrow T(n) = T(n/2) + \frac{1 \times \log n}{\log_2 a}$

$\frac{n \log_2 a}{\log_2 1}$

no

no

$equal \Rightarrow$

$TC = O(\log n)$

Time Complexity:

$$T(n) = T(n/2) + 1$$

$$O(\log n)$$

for all

Number of Multiplications :

1) if n is in the form of power of 2 =
 $m(n) = m(n/2) + 1$ $\Rightarrow \frac{\log_2 n}{\log_2 2}$
 $\log_2 n$

2) not in the form of 2 =
 $m(n) = m(n/2) + 2$

2n a^n $\Rightarrow$ how many mult $\Rightarrow$ ~~$\log n$~~

$\Rightarrow$ ~~$u \times u \times u \times u \times u$~~

~~$\Rightarrow a^n$~~

